Contract Risk Analysis via Context-Conditioned LLM Adaptation

Aarav Singh Luthra, Devyansh Choudhary, Saksham Bali, Uday Sodhi

Abstract

Legal contracts such as lease agreements, employment contracts, freelance agreements, and digital terms and conditions often contain clauses that impose financial or legal obligations that may not be immediately clear to non-expert users. In this project, we develop a contract analysis system that extends a pretrained instruction-tuned Large Language Model (LLM) with structured, clause-level risk extraction capabilities. The system first preprocesses uploaded contracts through deterministic document parsing and clause segmentation to produce semantically coherent clause units. The system then performs structured risk **classification** conditioned on user role and contract domain using parameter-efficient fine-tuning via Low-Rank Adaptation (LoRA). The model produces **categorical risk levels (Low, Medium, High)** along with concise natural-language explanations tailored to the specified user context. Optionally, we further integrate a Retrieval-Augmented Generation (RAG) module to support grounded cross-questioning over uploaded contracts. The system is evaluated using classification accuracy, precision, recall, macro F1 score, contextual sensitivity analysis, and human evaluation of generated explanations.

1 Introduction

Legal contracts are complex textual artifacts that encode rights, obligations, liabilities, and procedural constraints. Individuals frequently engage with employment agreements, lease contracts, freelance service agreements, non-disclosure agreements, and digital terms and conditions.

Interpreting clause-level implications in such documents requires careful attention to language that may signal risk depending on the reader's role and priorities.

Instruction-tuned Large Language Models (LLMs) provide strong general-purpose reasoning capabilities and can generate natural-language explanations for textual inputs. However, out-of-the-box models are not explicitly trained to produce structured, clause-level risk assessments conditioned on user role and contextual attributes.

In this project, we aim to adapt a pretrained instruction-tuned LLM to perform structured, context-conditioned legal risk extraction. The system:

1. Preprocesses uploaded contracts and segments them into coherent clauses.
2. Performs clause-level risk scoring conditioned on user role, contract domain, and risk tolerance.
3. produces structured outputs containing the predicted risk level and a concise explanation.
4. Optionally supports grounded cross-questioning via retrieval augmentation.

Our objective is to study how parameter-efficient adaptation and explicit context conditioning can enable a pretrained language model to internalize user-aware legal risk interpretation in a controlled and measurable manner.

2 Problem Formulation

Let a contract be represented as an ordered set of clauses:

$$C = \{c_1, c_2, \ldots, c_n\}$$

where each clause c_i is a semantically coherent unit extracted from a document belonging to domain D

(e.g., Lease, Employment, Freelance, NDA, Terms & Conditions).

Each clause is analyzed relative to a user context consisting of:

- User role U (e.g., tenant, employee, freelancer
- Contract domain D

2.1 Preprocessing and Clause Segmentation

Before modeling, uploaded documents undergo a preprocessing pipeline:

1. Document parsing (PDF/DOCX → raw text).
2. Text normalization (removal of headers, footers, page numbers, and formatting artifacts).
3. Deterministic clause boundary detection using rule-based heuristics such as: numbered headings (e.g., "1.", "2.1"), section indicators ("Section", "Article"), and capitalized or stylized headers.
4. Sentence-level coherence checks to merge fragmented segments and ensure semantically meaningful clause units.

This pipeline produces stable and reproducible clause-level inputs for modeling.

Let U denote structured user context consisting of: User Role (Tenant, Employer, Employee, Freelancer, Consumer) and Risk Tolerance (Low, Medium, High).

The core modeling objective is to learn:

$$f(c_i, U, D) \rightarrow (r_i, e_i)$$

where

- r_i is a categorical risk level for clause c_i
- e_i is a natural-language explanation describing the potential implication of the clause for the specified user context

The risk level belongs to one of three categories:

- **Low**
- **Medium**
- **High**

For interactive querying, we define:

$$g(C, U, Q) \rightarrow A$$

where Q is a user question and A is a grounded answer generated using retrieved clauses from C.

The system consists of two logical modules sharing a fine-tuned LLM core:

(1) Risk Flagging Module—structured clause-level risk extraction

(2) Cross-Questioning Module—retrieval-conditioned generation.

3 Related Work

3.1 Legal NLP

Legal NLP research has explored domain-adaptive pretraining, clause classification, and document-level inference. LegalBERT (Chalkidis et al., 2020) demonstrates benefits of legal-domain pretraining. The CUAD dataset (Hendrycks et al., 2021) provides clause-level annotations for commercial contracts. ContractNLI (Koreeda and Manning, 2021) frames contract reasoning as a natural language inference problem. These works provide datasets and modeling strategies relevant to legal text processing but do not explicitly study user-conditioned clause-level risk scoring.

3.2 Instruction-Tuned LLMs and Parameter-Efficient Adaptation

Instruction-tuned models such as LLaMA (Touvron et al., 2023) and Mistral (Jiang et al., 2023) demonstrate strong zero-shot reasoning capabilities. Low-Rank Adaptation (LoRA) (Hu et al., 2022) enables efficient fine-tuning by inserting low-rank adapter matrices into transformer layers while freezing base parameters. This approach allows domain-specific adaptation without full retraining.

3.3 Retrieval-Augmented Generation

Retrieval-Augmented Generation (Lewis et al., 2020) combines dense retrieval with language model generation to improve grounding and factual consistency. This framework can be applied to contract analysis to support clause-anchored question answering.

4 Methodology

4.1 Dataset Construction and Annotation

We construct a clause-level dataset from publicly available contracts across multiple domains: lease agreements, employment contracts, freelance agreements, non-disclosure agreements, and digital Terms & Conditions. Each contract is processed through the preprocessing and clause segmentation pipeline described earlier.

The primary dataset for this project is based on the CUAD dataset, which provides clause-level annotations for commercial contracts. However, CUAD does not include risk-level labels conditioned on user roles to the best of our knowledge, existing legal NLP datasets do not provide role-conditioned clause-level risk annotations, motivating the dataset construction process used in this project.

To address this limitation, we design a semi-automatic annotation pipeline for generating role-conditioned risk labels.

Annotation Pipeline

Rather than using an LLM arbitrarily as a label generator, we adopt a **structured three-step annotation pipeline** designed to produce reproducible and interpretable risk labels.

Step 1: Clause Extraction

Clause text is first extracted from CUAD documents using the preprocessing and clause segmentation pipeline described earlier. This process converts raw contracts into semantically coherent clause units that can be independently analyzed.

Step 2: LLM-Assisted Labelling Based on Risk Rubric

Initial candidate risk labels using a strong LLM (ChatGPT or similar models) based on a risk rubric defined by us.

Risk Rubric Definition

Before annotation begins, we define explicit criteria that determine how risk levels should be assigned based on both clause content and the user's role in the contract.
Examples include:

- A **unilateral termination clause** is considered **High risk for an Employee** but **Low risk for an Employer**.
- A clause requiring **tenants to bear all repair costs** is **High risk for Tenants** but **Low risk for Landlords**.

This rubric acts as the ground-truth definition for annotation and ensures that labels follow **deterministic criteria rather than subjective interpretation**.

LLM-Assisted Labelling

The LLM is prompted with the rubric and asked to classify each clause **strictly according to the defined criteria**, rather than relying on general legal reasoning.
Providing the rubric within the prompt ensures the model functions as a **semantic interpreter of predefined rules**, rather than an independent label generator.

Role Augmentation

To capture role-specific interpretations of contract clauses, the dataset is augmented by generating multiple training examples per clause, one for each relevant user role.
Example roles include:

- Tenant
- Employer
- Employee
- Freelancer
- Consumer

Not all roles apply to every clause. For example, a lease clause may be annotated for Tenant and Landlord roles, but not for Freelancer.

Step 3: Manual Verification

To ensure annotation quality, a random sample of approximately 10–15% of generated labels is manually reviewed by team members.
Annotations are checked against the predefined rubric to identify inconsistencies or systematic errors. Any discovered issues are corrected and the corrected examples are incorporated as few-shot examples to improve the consistency of the remaining labeling process.

Special attention is given to annotation consistency across user roles to ensure that role-conditioned differences are meaningful and systematically represented.

We target approximately 800–1500 clause-role examples for LoRA fine-tuning of a 7–8B parameter model. Dataset quality and internal consistency are central to this project, as the model's ability to learn context-conditioned behavior depends directly on supervision quality.

Because the CUAD annotation pipeline is currently in progress, we conduct initial baseline experiments using the Legal Indian Contract Clauses Dataset that contains:

- Clause text
- Clause type
- Risk level

This dataset does not contain user-role annotations and is therefore used only to obtain preliminary baseline results for clause-level risk classification. The same modelling pipeline will later be evaluated on the CUAD-based annotated dataset once labelling is completed.

4.2 Risk Schema

Each clause is assigned one of three categorical risk levels:

- **Low Risk:** Clause has minimal adverse implications for the specified user role.
- **Medium Risk:** Clause may introduce obligations or conditions requiring attention.
- **High Risk:** Clause introduces significant legal or financial exposure for the user.

Optional risk type tags (Financial, Termination, Liability, Ambiguity) are used for interpretability but are not core training targets.

4.3 Base Model and LoRA Fine-Tuning

We use a pretrained instruction-tuned LLM (7–8B parameters) as the base model. Fine-tuning is performed using QLoRA with 4-bit quantization for memory efficiency, frozen base weights, and trainable low-rank adapters inserted into attention projection layers.

Training examples follow a structured instruction format:

User Role: [role] | Domain: [domain] | Risk Tolerance: [tolerance]

Clause: [text]
Output:
Risk Level: [label]
Explanation: [text]

Cross-entropy loss is computed over output tokens only. This enables the model to learn structured, context-aware risk extraction without architectural modification.

4.4 Context Conditioning

User role and domain are injected explicitly during both training and inference. This encourages learning of the joint mapping: Risk = f(Clause, Role, Domain). Contextual sensitivity is evaluated by presenting identical clauses under different user roles and measuring variation in predicted outputs.

4.5 Risk Flagging Module

For inference, the pipeline operates as follows:

1. The user uploads a contract
2. The contract is segmented into clauses
3. Each clause is passed to the fine-tuned LLM with user context
4. Structured outputs are parsed and stored
5. Clauses are ranked by predicted risk.

4.6 Cross-Questioning Module (Optional RAG)

If implemented: clause embeddings are computed using a pretrained sentence encoder and stored in a FAISS index. For each user query, top-k clauses are retrieved. Retrieved clauses and user context are passed to the LLM to generate a grounded response.

4.7 Planned Experiments

To evaluate the effectiveness of the proposed system, we plan the following experiments:

Experiment 1: Zero-Shot Baseline

We evaluate **LLaMA 3.1 8B Instruct** and **Mistral 7B Instruct** in a zero-shot setting without any fine-tuning or role context.
The model receives only:

- Clause text
- Clause type

This experiment establishes the **baseline performance** and identifies which model demonstrates stronger out-of-the-box legal reasoning ability.

Experiment 2: Fine-Tuned Baseline (No Role Context)

Both LLaMA 3.1 8B and Mistral 7B Instruct are fine-tuned using QLoRA on the clause dataset **without user role information**.
This isolates the contribution of **domain adaptation alone**.
The better-performing model is selected for the next experiment.

Experiment 3: Role-Conditioned Fine-Tuning
The selected model from Experiment 2 is fine-tuned on the **role-augmented dataset** where each clause appears with multiple user roles.
This experiment evaluates whether explicitly conditioning on user role improves structured risk prediction.

5 Evaluation

5.1 Proposed Evaluation Metrics

Structured Risk Prediction

Categorical risk level outputs are parsed and evaluated using Accuracy, Precision, Recall, and Macro F1-score.

Contextual Sensitivity

For identical clauses evaluated under different roles: variance in predicted risk scores and comparison between role-conditioned and role-agnostic models.

RAG Evaluation

Grounding rate (overlap with retrieved clauses) and hallucination rate (human evaluation).

Human Evaluation

100 clause explanations are evaluated on Clarity, Role Relevance, and Perceived Correctness using a 1–5 Likert scale.

5.2 Baseline Results

Model	Accuracy	Macro F1
Zero-shot LLaMA 3.1 8B	51.5%	0.446
Zero-shot Mistral 7B Instruct	52.0%	0.428
Fine-tuned LLaMA (no role)	TBD	TBD
Fine-tuned Mistral (no role)	TBD	TBD
Ours - best model (role-conditioned)	TBD	TBD

We evaluate two models in a zero-shot setting - LLaMA 3.1 8B and Mistral 7B Instruct - on a stratified sample of 200 clauses. With three risk classes, a random baseline would yield 33.3% accuracy; both models achieve approximately **51–52% accuracy**, indicating that general-purpose LLMs possess limited capability for legal risk classification but still require domain-specific adaptation for reliable performance. We select LLaMA 3.1 8B for further fine-tuning experiments based on its slightly higher Macro F1 score (0.446 vs 0.428).

5.3 Analysis and Discussion

Dataset Construction and Annotation Strategy

The primary concern with dataset construction was ensuring that risk labels reflect a consistent and reproducible definition of risk rather than the internal biases of whatever model generates them. General-purpose LLMs are not explicitly trained to perform structured clause-level legal risk classification, they assign risk based on patterns learned during pretraining which are inconsistent, opaque, and not grounded in any explicit standard. To address this, we first define a risk rubric ourselves that specifies exact criteria for what constitutes Low, Medium, and High risk for each user role, and then constrain the LLM strictly to applying these criteria. This reduces the LLM's role to rule application rather than rule invention, ensuring that labels are traceable back to our own definitions. Manual verification of a subset further catches cases where even rule guided annotation produces inconsistent outputs, particularly on ambiguous clauses where legal language is deliberately vague.

Model Selection

LLaMA 3.1 8B and Mistral 7B Instruct were shortlisted ‘because they are the two strongest openly available instruction tuned models at the 7 to 8 billion parameter scale, making them realistic candidates for fine tuning without requiring large scale compute infrastructure. A key decision before fine tuning was which of these to commit to, since running both through the full pipeline was not practical. The zero shot evaluation was designed specifically to resolve this by measuring both accuracy and Macro F1 on the same sample.

Importance of Role Conditioning

The zero shot results exposed the most fundamental limitation of a general-purpose model for this task, it assigns identical risk to identical clauses regardless of user role. A termination clause is not inherently risky, it is risky for the employee and routine for the employer, and no amount of domain fine tuning alone can capture this asymmetry because the risk is not in

the clause text but in the relationship between the clause and the user. The role conditioned fine tuning experiment directly addresses this by explicitly injecting user role into the input, and the performance gap between the role agnostic and role conditioned models will determine whether this intervention is sufficient.

Limitations

This document describes a proposed system and experimental design. Evaluation results, ablation studies, and cross-domain generalization analyses are deferred to the full experimental report.

6. Conclusion

In this project, we propose a framework for context-conditioned contract risk analysis using instruction-tuned large language models.

The system performs clause-level risk classification conditioned on user role and contract domain, producing structured outputs consisting of categorical risk levels and natural-language explanations.

Our methodology combines deterministic contract preprocessing, parameter-efficient model adaptation using QLoRA, and a structured annotation pipeline guided by a predefined risk rubric. Initial baseline experiments comparing LLaMA 3.1 8B and Mistral 7B demonstrate that general-purpose language models possess limited zero-shot capability for legal risk classification but benefit significantly from domain adaptation.

Based on the zero-shot evaluation results, LLaMA 3.1 8B demonstrated slightly stronger performance on the training dataset and has therefore been selected as the base model for the subsequent fine-tuning experiments.

The ongoing annotation of the CUAD dataset will enable the next stage of experiments involving role-conditioned fine-tuning. These experiments will evaluate whether incorporating explicit user-role context improves clause-level risk prediction.

An area of ongoing exploration in our pipeline is whether to group semantically related clauses before classification, and if so, how. Two approaches are under consideration.

The first is **retrieval-based grouping**, where for each clause, the most similar clauses are retrieved and passed together as context. This is flexible and keeps risk attribution at the clause level, but may pull in clauses that are superficially similar rather than truly related in meaning.

The second is **cluster-based grouping**, where all clauses are partitioned into thematic groups first, and each group is classified together. This can surface risks that only emerge when multiple related clauses are read together, but produces a single risk label for the entire group, making it harder to pinpoint exactly which clause is problematic.

Both approaches have merit and the trade-offs are still being evaluated. Clause grouping is planned as a future enhancement to the current pipeline once the core classification system is fully benchmarked.

Overall, this work aims to demonstrate how structured context conditioning can enable LLMs to perform interpretable and user-aware legal risk analysis.

References

Ilias Chalkidis, Manos Fergadiotis, Prodromos Malakasiotis, Nikolaos Aletras, and Ion Androutsopoulos. 2020. LEGAL-BERT: The muppets straight out of law school. In Findings of EMNLP.

Dan Hendrycks, Collin Burns, Anya Chen, and Spencer Ball. 2021. CUAD: An expert-annotated NLP dataset for legal contract review. In NeurIPS Datasets and Benchmarks.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In ICLR.

Yuta Koreeda and Christopher Manning. 2021. ContractNLI: A dataset for document-level natural language inference for contracts. In Findings of EMNLP.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, et al. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In NeurIPS.

Hugo Touvron, Thibaut Lavril, Gautier Izacard, et al. 2023. LLaMA: Open and efficient foundation language models. arXiv:2302.13971.

Individual Contributions

Aarav Singh Luthra

Aarav built and ran the zero-shot evaluation pipeline for both LLaMA 3.1 8B and Mistral 7B Instruct, producing the baseline performance metrics that guided model selection and determined which model the team would carry forward into fine-tuning.

Devyansh Choudhary

Designed the role-dependent risk rubric that defines how risk levels are assigned across different user roles and contract types, forming the conceptual foundation that the annotation pipeline and fine-tuning approach are built on.

Saksham Bali

Saksham developed the document preprocessing and clause segmentation pipeline that converts raw legal contracts into structured, semantically coherent clause units, ensuring clean and consistent inputs for both the annotation process and model training.

Uday Sodhi

Uday designed and implemented the CUAD annotation pipeline, including the rubric-based prompting strategy and role augmentation workflow that generates role-conditioned training examples across multiple user roles and contract domains, forming the dataset that the fine-tuned model will be trained on.

Project GitHub Repository:

https://github.com/Saksham-Bali/Advanced_NLP_proj